Day 5: First Wireshark Capture (Loopback Ping)

Level 1, Month 1, Networking foundations

What I learned

- Wireshark lists the network adapters of my computer. With no connection, the useful one is `Adapter for loopback traffic capture`.
- `127.0.0.1` is the loopback address: the computer talks to itself, so source and destination are the same.
- A ping is a pair of ICMP packets: an Echo request, then an Echo reply. The `Info` column links them, for example `(reply in 2)`.
- `seq` increases by one in each request (9, 10, 11, 12).
- The reply comes in about 0.00007 seconds because the packet never leaves the machine.

Lab: my first capture

1. Opened the loopback adapter in Wireshark.

2. Applied the display filter `icmp`.
3. Ran `ping 127.0.0.1` in Command Prompt.
4. Result: 8 packets, 4 requests and 4 replies, all from `127.0.0.1` to `127.0.0.1`.

Capture file: `day-05-loopback-ping.pcapng` (only loopback traffic).

Reading an IPv4 header

Field	Value	Meaning
Version	4 (`0100`)	IPv4
Header Length	20 bytes (5)	5 x 4 bytes
Total Length	60	Whole packet: IP header + ICMP message
Time to Live	128	Goes down by 1 at each router, packet dropped at 0
Protocol	ICMP (1)	What is inside the packet

Field	Value	Meaning
Source Address	127.0.0.1	4 bytes = 32 bits
Destination Address	127.0.0.1	4 bytes = 32 bits
Header Checksum	0x0000 (validation disabled)	Normal here, the packet never left the machine

Check questions I answered correctly:

- `0100` = 4, `0101` = 5, `0111` = 7.
- 4 bytes x 8 = 32 bits.
- Total length 60 - IP header 20 = 40 bytes for the ICMP message.

Warm-up: 172.31.203.130/21

Item	Value
Bits d'hote	11
Taille du bloc	8
Adresse reseau	172.31.200.0
Premiere adresse	172.31.200.1

Item	Value
Broadcast	172.31.207.255
Derniere adresse	172.31.207.254
Hotes valides	2046

Error log

Mistake	Correct answer	Why
Taille du bloc 16 for /21	8	I used /20 instead of /21. 24 - 21 = 3, then 2^3 = 8.
Premiere 172.31.193.1	172.31.200.1	First host changes only the fourth number.
Derniere equal to broadcast (.207.255)	.207.254	Derniere is one before broadcast, so it must be smaller.

Mistake	Correct answer	Why
Valid hosts 2048	2046	2048 is all the addresses. Subtract 2.

Main lessons

- Use the mask of the problem, not the one from the previous problem.
- Derniere is always smaller than broadcast. Check it before sending.
- Hotes valides = 2^(bits d'hote) - 2. I forgot the "- 2" again, so it needs a card.
- I can already read real packets: the header fields match what I learned in the first sessions.

Anki

Card for the rule `Hotes valides = 2^(bits d'hote) - 2`, and cards for TTL, Protocol and Header Length.

Next session

Filter by a MAC address and look at an ARP packet,
and a short subnetting review.